CPSC-354 Report

Mitchell Toney
Chapman University

November 23, 2025

Abstract

Contents

1 Introduction

2 Week by Week

2.1 Week 1: HW1

The MU puzzle is a puzzle created by Douglas Hofstadter. It consists of four rules that can be applied to a string MI.

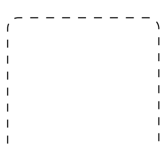
1. $xI \to xIU$
2. $Mx \to Mxx$
3. $xIIIy \to xUy$
4. $xUUy \to xy$

When first approaching this puzzle, the first strategy that came to mind was to take advantage of rule number 2 to keep duplicating the I's until there is a multiple of three, then using rules 3 and 4 to get rid of the I's and leave a remaining U.

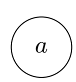
The issue with this is that $2^n \bmod 3$ will never equal 0, it infinitely cycles between equaling 1 and 2, and without being able to get rid of all the I's, which would require them being a multiple of 3, you will never be able to get MU.

Thus, the puzzle is not solvable.

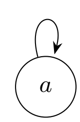
2.2 Week 2: HW2



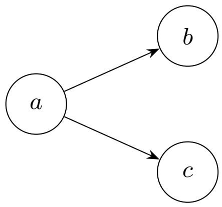
1. $A = \varnothing,\ R = \varnothing$



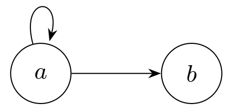
2. $A = \{a\},\ R = \varnothing$



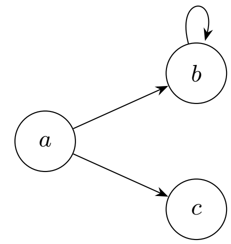
3. $A = \{a\},\ R = \{(a, a)\}$



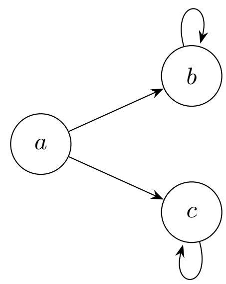
4. $A = \{a, b, c\}$, $R = \{(a, b), (a, c)\}$



5. $A = \{a, b\}$, $R = \{(a, a), (a, b)\}$



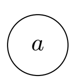
6. $A = \{a, b, c\}$, $R = \{(a, b), (b, b), (a, c)\}$



7. $A = \{a, b, c\}$, $R = \{(a, b), (b, b), (a, c), (c, c)\}$

#	Terminating	Confluent	Unique NFs
1	Yes	Yes	Yes
2	Yes	Yes	Yes
3	No	Yes	No
4	Yes	No	No
5	No	Yes	Yes
6	No	No	No
7	No	No	No

Confluent True, Terminating True, Unique NFs True



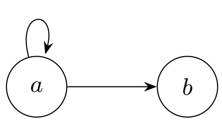
$A = \{a\}$, $R = \varnothing$

Confluent True, Terminating True, Unique NFs False

impossible

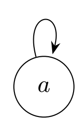
(no ARS exists)

Confluent True, Terminating False, Unique NFs True



$A = \{a, b\}$, $R = \{(a, a), (a, b)\}$

Confluent True, Terminating False, Unique NFs False



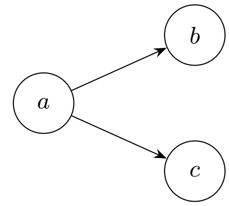
$A = \{a\}$, $R = \{(a, a)\}$

Confluent False, Terminating True, Unique NFs True

impossible

(no ARS exists)

Confluent False, Terminating True, Unique NFs False



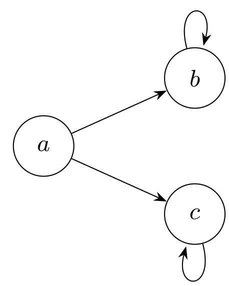
$A = \{a, b, c\},\ R = \{(a, b), (a, c)\}$

Confluent False, Terminating False, Unique NFs True

impossible

(no ARS exists)

Confluent False, Terminating False, Unique NFs False



$A = \{a, b, c\},\ R = \{(a, b), (a, c), (b, b), (c, c)\}$

2.3 Week 3: HW3

2.3.1 Exercise 5

Consider rewrite rules:

$$\begin{aligned} ab &\to ba \\ ba &\to ab \\ aa &\to \\ b &\to \end{aligned}$$

Example Reductions

Reducing `abba`:

$$\begin{aligned} abba &\to baba \quad (\text{using } ab \to ba) \\ baba &\to bbaa \quad (\text{using } ba \to ab) \\ bbaa &\to baa \quad (\text{using } b \to \varepsilon) \\ baa &\to aba \quad (\text{using } ba \to ab) \\ aba &\to baa \quad (\text{using } ab \to ba) \end{aligned}$$

There is an infinite loop between `aba` and `baa`.

Reducing `bababa`:

$$\begin{aligned} bababa &\to ababab \quad (\text{using } ba \to ab) \\ ababab &\to baabab \quad (\text{using } ab \to ba) \\ baabab &\to ababab \quad (\text{using } ba \to ab) \end{aligned}$$

This is an infinite loop between `ababab` and `baabab`.

Why the ARS is not terminating The ARS is not terminating because the rules $ab \to ba$ and $ba \to ab$ create infinite cycles. These rules allow us to swap adjacent a and b characters indefinitely, leading to non-terminating reduction sequences.

Non-equivalent strings Two strings that are not equivalent: `a` and `aa`.

The string `a` cannot be reduced further, while `aa` reduces to nothing using the rule $aa \to$. Since *nothing* $\neq a$, these strings are in different equivalence classes.

Equivalence classes The equivalence relation $\leftrightarrow^*$ has infinitely many equivalence classes. Each equivalence class can be characterized by the number of a's modulo 2 and the number of b's modulo 1.

The equivalence classes are:

- $[\varepsilon]$: strings with even number of a's and no b's
- $[a]$: strings with odd number of a's and no b's
- $[b]$: strings with any number of a's and at least one b

The normal forms are: ε, a, and b.

Modifying the ARS to be terminating To make the ARS terminating without changing equivalence classes, we can remove the symmetric rules and keep only one direction:

$$\begin{aligned} ba &\to ab \\ aa &\to \varepsilon \\ b &\to \varepsilon \end{aligned}$$

This eliminates the infinite cycles while preserving the same equivalence relation.

2.3.2 Semantic question

Parity of a's: "Does this string contain an odd number of a's?"
Answer: Yes if the normal form is a, No if the normal form is ε.

Exercise 5b

Consider rewrite rules:

$$\begin{aligned} ab &\to ba \\ ba &\to ab \\ aa &\to a \\ b &\to \end{aligned}$$

Reducing `abba`:

$$\begin{aligned} abba &\to baba \\ baba &\to abba \end{aligned}$$

Reducing `bababa`:

$$\begin{aligned} bababa &\to ababab \\ ababab &\to bababa \end{aligned}$$

Why not terminating. The symmetric swaps $ab \leftrightarrow ba$ allow infinite rewriting.

Non-equivalent strings. a and ε are not equivalent.

Equivalence classes. Exactly two: $[\varepsilon]$ (no a's; all b's delete) and $[a]$ (at least one a; since $aa \sim a$). Normal forms (under a terminating orientation): ε and a.

Terminating variant.

$$
\begin{aligned}
ba &\to ab \\
aa &\to a \\
b &\to
\end{aligned}
$$

This gives a complete semantics to the ARS: it computes the invariant for any input string.

2.4 Week 4: HW4

2.4.1 HW4.1: Termination Proof for GCD

Consider the following algorithm:

```
while b != 0:
    temp = b
    b = a mod b
    a = temp
return a
```

Assume: Work over integers with Euclidean division: inputs $a \in \mathbb{Z}$, $b \in \mathbb{N}$; if $b \neq 0$ then $0 \leq a \bmod b < b$.

Model: States $A = \mathbb{Z} \times \mathbb{N}$. One step $(a, b) \to (a', b')$ is one loop iteration.

Measure: $\phi : A \to \mathbb{N},\ \phi(a, b) = b$.

Show: $(a, b) \to (a', b') \Rightarrow \phi(a', b') < \phi(a, b)$.

If $b \neq 0$, the update sets $b' = a \bmod b$ with $0 \leq b' < b$; hence $\phi(a', b') = b' < b = \phi(a, b)$.

ϕ strictly decreases in $\mathbb{N}$, so there is no infinite $\to$-chain; eventually $b = 0$ and the loop stops. Thus the algorithm terminates under the stated conditions.

2.4.2 HW4.2: Termination Proof for Merge Sort

Consider the following fragment of an implementation of merge sort:

```
function merge_sort(arr, left, right):
    if left >= right:
        return
    mid = (left + right) / 2
    merge_sort(arr, left, mid)
    merge_sort(arr, mid+1, right)
    merge(arr, left, mid, right)
```

Prove that

$\phi(\texttt{left}, \texttt{right}) = \texttt{right} - \texttt{left} + 1$

is a measure function for merge_sort.

Show: For

```
merge_sort(arr, left, right):
  if left >= right: return
    mid = floor((left + right)/2)
  merge_sort(arr, left, mid)
  merge_sort(arr, mid+1, right)
  merge(arr, left, mid, right)
```

the function $\phi(\text{left}, \text{right}) = \text{right} - \text{left} + 1$ is a strictly decreasing measure, hence `merge_sort` terminates.

Assume: `left,right`$\in \mathbb{Z}$ with $0 \leq \text{left} \leq \text{right} < |arr|$. Division for `mid` is integer. If `left < right`, then $\text{left} \leq \text{mid} < \text{right}$. `merge` makes no recursive calls.

Model: States are intervals $A = \{(l, r) \in \mathbb{Z}^2 \mid l \leq r\}$. A "step" is a recursive edge from (l, r) (with $l < r$) to each child (l, mid) and $(\text{mid} + 1, r)$.

Measure: $\phi : A \to \mathbb{N}, \quad \phi(l, r) = r - l + 1.$

Let $l < r$ and $\text{mid} = \lfloor (l + r)/2 \rfloor$ so $l \leq \text{mid} < r$.

First child: $\phi(l, \text{mid}) = \text{mid} - l + 1 \leq (r - 1) - l + 1 = r - l = \phi(l, r) - 1 < \phi(l, r)$.

Second child: $\phi(\text{mid} + 1, r) = r - \text{mid} \leq r - l = \phi(l, r) - 1 < \phi(l, r)$.

Every recursive edge strictly decreases ϕ in $\mathbb{N}$, which is well-founded. Thus no infinite recursion is possible; the calls bottom out at states with $\phi \in \{0, 1\}$ (i.e., `left` $\geq$ `right`), where the function returns. Therefore $\phi(\text{left}, \text{right}) = \text{right} - \text{left} + 1$ is a valid measure and `merge_sort` terminates under the stated conditions.

2.5 Week 5: HW5

2.5.1 Lambda Calculus Workout Evaluation (Corrected)

Evaluate: $(\lambda f.\lambda x.\, f(f\, x))(\lambda g.\lambda y.\, g(g(g\, y)))$

Use α-renaming to avoid capture.

$$
\begin{aligned}
(\lambda f.\lambda x.\, f(f\, x))(\lambda g.\lambda y.\, g(g(g\, y))) &\to \lambda x.\, (\lambda g.\lambda y.\, g(g(g\, y)))\big((\lambda g.\lambda y.\, g(g(g\, y)))\, x\big) \\
(\lambda g.\lambda y.\, g(g(g\, y)))\, x &\to \lambda y.\, x(x(x\, y)) \quad (\alpha\text{-rename inner } y) \\
\Rightarrow\ \lambda x.\, (\lambda g.\lambda y.\, g(g(g\, y)))\, (\lambda y.\, x(x(x\, y))) &\to \lambda x.\lambda y.\, x^9 y.
\end{aligned}
$$

Hence the normal form is $\lambda f.\lambda x.\, f^9 x$ (Church numeral 9).

2.6 Week 6: HW6

2.6.1 Fixed Point Combinator Exercise

Compute `fact 3` following the computation rules for `fix`, `let`, and `let rec`:

Given:

$$E_0 \ = \ \mathsf{let\ rec\ fact} = \lambda n.\ \mathsf{if}\ n = 0\ \mathsf{then}\ 1\ \mathsf{else}\ n * \mathsf{fact}\,(n-1)\ \mathsf{in\ fact}\ 3$$

Abbreviations:

$$\mathsf{F} \ \stackrel{\text{def}}{=}\ \lambda f.\, \lambda n.\ \mathsf{if}\ n = 0\ \mathsf{then}\ 1\ \mathsf{else}\ n * f\,(n-1) \qquad\qquad \mathsf{FACT} \ \stackrel{\text{def}}{=}\ \mathsf{fix}\,\mathsf{F}$$

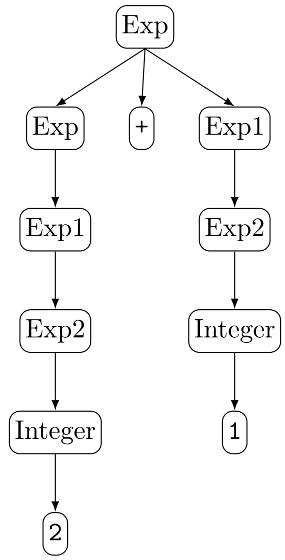
Computation:

$$
\begin{aligned}
E_0 &\xrightarrow{\text{def of let rec}} \mathsf{let\ fact} = \mathsf{fix}(\lambda f.\lambda n.\mathsf{if}\ n = 0\ \mathsf{then}\ 1\ \mathsf{else}\ n * f\,(n-1))\ \mathsf{in\ fact}\,3 \\
&\xrightarrow{\text{def of let}} (\lambda \mathsf{fact}.\ \mathsf{fact}\,3)\ \mathsf{fix}(\lambda f.\lambda n.\mathsf{if}\ n = 0\ \mathsf{then}\ 1\ \mathsf{else}\ n * f\,(n-1)) \\
&\xrightarrow{\beta} \big(\mathsf{fix}(\lambda f.\lambda n.\mathsf{if}\ n = 0\ \mathsf{then}\ 1\ \mathsf{else}\ n * f\,(n-1))\big)\ 3 \\
&\equiv \mathsf{FACT}\,3 \\
&\xrightarrow{\text{def of fix}} (\mathsf{F\,FACT})\ 3 \\
&\xrightarrow{\beta} (\lambda n.\ \mathsf{if}\ n = 0\ \mathsf{then}\ 1\ \mathsf{else}\ n * \mathsf{FACT}\,(n-1))\ 3 \\
&\xrightarrow{\beta} \mathsf{if}\ 3 = 0\ \mathsf{then}\ 1\ \mathsf{else}\ 3 * \mathsf{FACT}\,(3-1) \\
&\xrightarrow{\text{def of if}} 3 * \mathsf{FACT}\,(3-1) \\
&\xrightarrow{\text{arith}} 3 * \mathsf{FACT}\,2 \\
&\xrightarrow{\text{def of fix}} 3 * (\mathsf{F\,FACT})\ 2 \\
&\xrightarrow{\beta} 3 * (\lambda n.\ \mathsf{if}\ n = 0\ \mathsf{then}\ 1\ \mathsf{else}\ n * \mathsf{FACT}\,(n-1))\ 2 \\
&\xrightarrow{\beta} 3 * \mathsf{if}\ 2 = 0\ \mathsf{then}\ 1\ \mathsf{else}\ 2 * \mathsf{FACT}\,(2-1) \\
&\xrightarrow{\text{def of if}} 3 * (2 * \mathsf{FACT}\,(2-1)) \\
&\xrightarrow{\text{arith}} 3 * (2 * \mathsf{FACT}\,1) \\
&\xrightarrow{\text{def of fix}} 3 * \big(2 * (\mathsf{F\,FACT})\ 1\big) \\
&\xrightarrow{\beta} 3 * \big(2 * (\lambda n.\ \mathsf{if}\ n = 0\ \mathsf{then}\ 1\ \mathsf{else}\ n * \mathsf{FACT}\,(n-1))\ 1\big) \\
&\xrightarrow{\beta} 3 * \big(2 * \mathsf{if}\ 1 = 0\ \mathsf{then}\ 1\ \mathsf{else}\ 1 * \mathsf{FACT}\,(1-1)\big) \\
&\xrightarrow{\text{def of if}} 3 * \big(2 * (1 * \mathsf{FACT}\,(1-1))\big) \\
&\xrightarrow{\text{arith}} 3 * \big(2 * (1 * \mathsf{FACT}\,0)\big) \\
&\xrightarrow{\text{def of fix}} 3 * \big(2 * (1 * (\mathsf{F\,FACT})\ 0)\big) \\
&\xrightarrow{\beta} 3 * \big(2 * (1 * (\lambda n.\ \mathsf{if}\ n = 0\ \mathsf{then}\ 1\ \mathsf{else}\ n * \mathsf{FACT}\,(n-1))\ 0)\big) \\
&\xrightarrow{\beta} 3 * \big(2 * (1 * \mathsf{if}\ 0 = 0\ \mathsf{then}\ 1\ \mathsf{else}\ 0 * \mathsf{FACT}\,(0-1))\big) \\
&\xrightarrow{\text{def of if}} 3 * \big(2 * (1 * 1)\big) \\
&\xrightarrow{\text{arith}} 3 * (2 * 1) \xrightarrow{\text{arith}} 3 * 2 \xrightarrow{\text{arith}} 6\ .
\end{aligned}
$$

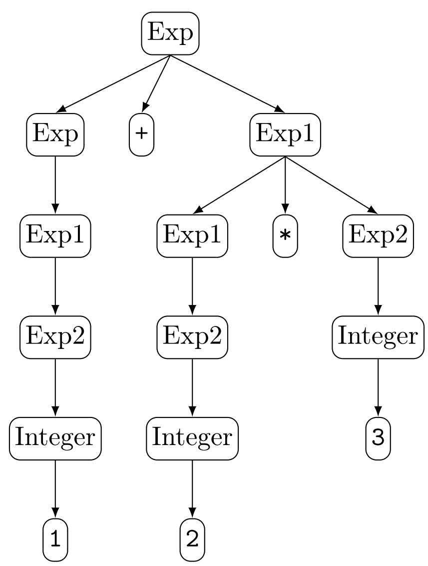
Result: `fact 3 = 6`

2.7 Week 7: HW 7 - Parse Trees Exercise

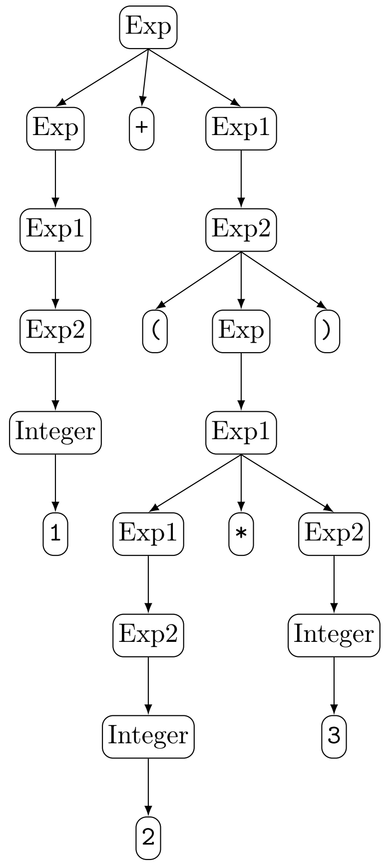
2.7.1 String: $2 + 1$



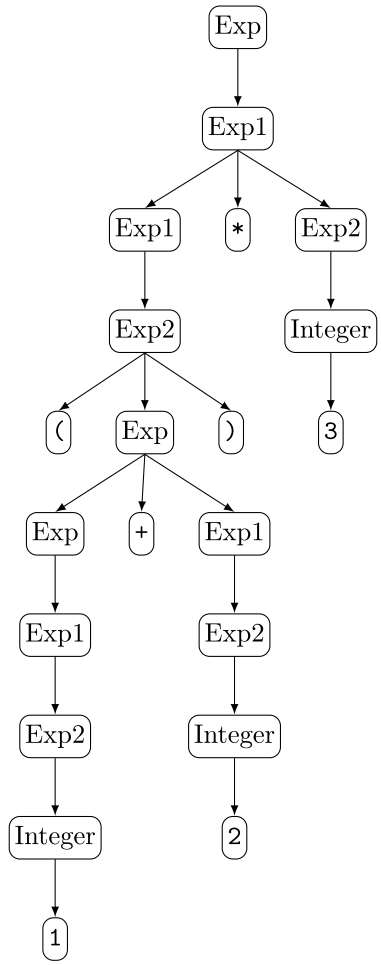
2.7.2 String: $1 + 2 * 3$



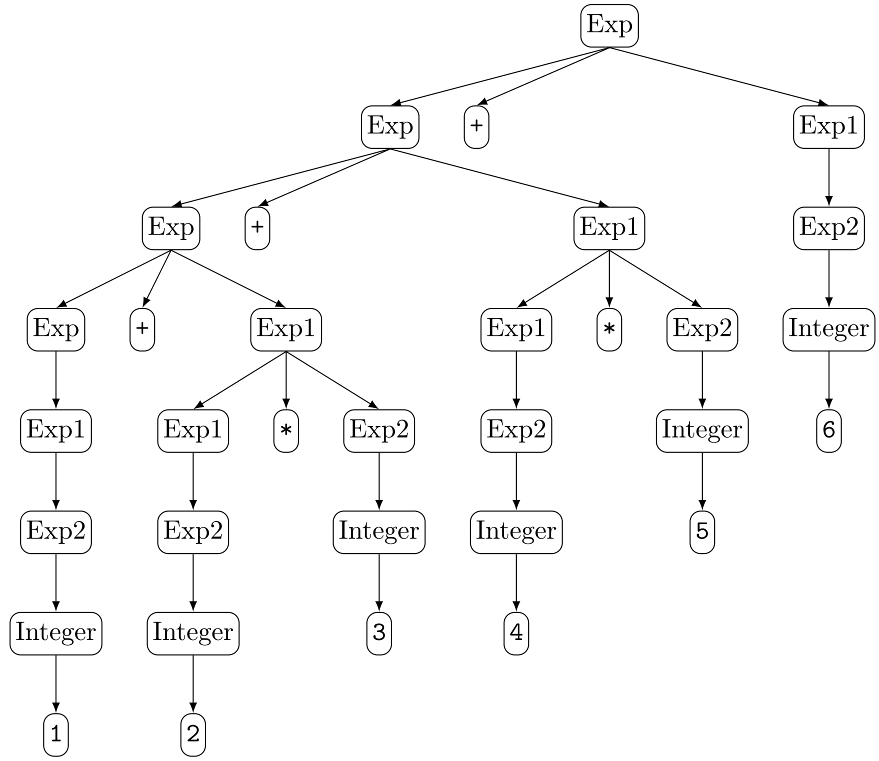
2.7.3 String: $1 + (2 * 3)$



2.7.4 String: $(1 + 2) * 3$



2.7.5 String: $1 + 2 * 3 + 4 * 5 + 6$



2.8 Week 8: HW 8

2.8.1 Exercise 1

Level 5:

$a + (b + 0) + (c + 0) = a + b + c$

```
rw [add_zero]
rw [add_zero]
rfl
```

Level 6:

$a + (b + 0) + (c + 0) = a + b + c$

```
rw [add_zero c]
rw [add_zero b]
rfl
```

Level 7:

$succn = n + 1$

```
rw [one_eq_succ_zero]
rw [add_succ]
rw[add_zero]
rfl
```

Level 8:

$2 + 2 = 4$

```
rw [two_eq_succ_one]
rw [four_eq_succ_three]
rw [three_eq_succ_two]
rw [two_eq_succ_one]
rw [one_eq_succ_zero]
rw [add_succ]
rw [add_succ]
rw [add_zero]
rfl
```

Level 7 Natural Language Proof:

$$\operatorname{succ} n = n + 1$$

Replace 1 by $\operatorname{succ} 0$:

$$\operatorname{succ} n = n + \operatorname{succ} 0$$

Use the definition of addition with a successor:

$$\operatorname{succ} n = \operatorname{succ}(n + 0)$$

Simplify $n + 0$ to n:

$$\operatorname{succ} n = \operatorname{succ} n$$

Both sides are identical, so the equality holds.

2.9 Week 9: HW 9

2.9.1 No Induction

```
rw [add_assoc a b c]
rw [add_comm b c]
rw [← add_assoc a c b]
```

For all $(a, b, c \in \mathbb{N})$,

$$a + b + c = a + c + b.$$

$$\begin{aligned} a + b + c &= a + (b + c) \quad &&\text{(associativity of } (+)\text{)} \\ &= a + (c + b) \quad &&\text{(commutativity of } (+)\text{)} \\ &= a + c + b \quad &&\text{(associativity of } (+)\text{)}. \end{aligned}$$

2.9.2 Induction

```
induction c with d hd
rw [add_zero, add_zero]
rfl

rw[add_succ]
rw[add_succ]
rw[succ_add]
rw[hd]
```

For all $(a, b, c \in \mathbb{N})$,

$$a + b + c = a + c + b.$$

By induction on c.

Base case ($c = 0$).

$$\begin{aligned} a + b + 0 &= a + b && \text{(by } (x + 0 = x)\text{)} \\ &= a + 0 + b && \text{(by } (x + 0 = x)\text{)}, \end{aligned}$$

Assume $a + b + d = a + d + b$:

$$\begin{aligned} a + b + (d + 1) &= (a + b + d) + 1 && \text{(by } (x + (y + 1) = (x + y) + 1)\text{)} \\ &= (a + d + b) + 1 && \text{(induction hypothesis)} \\ &= (a + d + 1) + b && \text{(by } ((x + y) + 1 = (x + 1) + y)\text{)} \\ &= a + (d + 1) + b && \text{(associativity)}. \end{aligned}$$

Thus $a + b + c = a + c + b$ for all $c \in \mathbb{N}$.

2.10 Week 10: HW 10

2.10.1 Level 6: Curry

Objects: C, D, S : Prop

Assumptions: $h : C \wedge D \to S$

Goal:

$C \to D \to S$

Solution:

```
exact λ c d ↦ h (and_intro c d)
```

2.10.2 Level 7: Un-Curry

Objects: C, D, S : Prop

Assumptions: $h : C \to D \to S$

Goal:

$C \wedge D \to S$

Solution:

```
exact λ cd ↦ h (and_left cd) (and_right cd)
```

2.10.3 Level 8: Go buy chips and dip!

Objects: C, D, S : Prop

Assumptions: $h : (S \to C) \wedge (S \to D)$

Goal:

$S \to C \wedge D$

Solution:

```
exact λ s ↦ and_intro (and_left h s) (and_right h s)
```

2.10.4 Level 9: Uncertain Snacks

Objects: R, S : Prop

Goal:

$R \to (S \to R) \wedge (\neg S \to R)$

Solution:

```
exact λ r ↦ and_intro (λ s ↦ r) (λ _ ↦ r)
```

Question: Does currying versus uncurrying ever change definitional equality enough to affect the rewrites from the homework?

2.11 Week 11: HW 11

2.11.1 Level 9: Allergy #1

Objects: P, A : Prop

Assumptions: $h : P \to \neg A$

Goal:

$\neg(P \wedge A)$

Solution:

```
exact λ pa : P ∧ A ↦ h pa.left pa.right
```

2.11.2 Level 10: Allergy #2

Objects: P, A : Prop

Assumptions: $h : \neg(P \wedge A)$

Goal:

$P \to \neg A$

Solution:

```
exact λ p : P ↦ λ a : A ↦ h (and_intro p a)
```

2.11.3 Level 11: Allergy: Triple Confusion

Objects: A : Prop

Assumptions: $h : \neg\neg\neg A$

Goal:

$\neg A$

Solution:

```
exact λ a : A ↦ h (λ na : ¬A ↦ na a)
```

2.11.4 Level 12

Objects: B, C : Prop

Assumptions: $h : \neg(B \to C)$

Goal:

$\neg\neg B$

Solution:

```
exact λ nb ↦ h (nb ≫ false_elim)
```

Question: How would the allergy scenario look different if explosion was rejected?

2.12 Week 12: HW 12

2.12.1 Completed execution

Execution for `hanoi 5 0 2`:

```
hanoi 5 0 2
    hanoi 4 0 1
        hanoi 3 0 2
            hanoi 2 0 1
                hanoi 1 0 2 = move 0 2
                move 0 1
                hanoi 1 2 1 = move 2 1
            move 0 2
            hanoi 2 1 2
                hanoi 1 1 0 = move 1 0
                move 1 2
                hanoi 1 0 2 = move 0 2
        move 0 1
        hanoi 3 2 1
            hanoi 2 2 0
                hanoi 1 2 1 = move 2 1
                move 2 0
                hanoi 1 1 0 = move 1 0
            move 2 1
            hanoi 2 0 1
                hanoi 1 0 2 = move 0 2
                move 0 1
                hanoi 1 2 1 = move 2 1
    move 0 2
    hanoi 4 1 2
        hanoi 3 1 0
            hanoi 2 1 2
                hanoi 1 1 0 = move 1 0
                move 1 2
                hanoi 1 0 2 = move 0 2
            move 1 0
            hanoi 2 2 0
                hanoi 1 2 1 = move 2 1
                move 2 0
                hanoi 1 1 0 = move 1 0
```

```
move 1 2
hanoi 3 0 2
    hanoi 2 0 1
        hanoi 1 0 2 = move 0 2
        move 0 1
        hanoi 1 2 1 = move 2 1
    move 0 2
    hanoi 2 1 2
        hanoi 1 1 0 = move 1 0
        move 1 2
        hanoi 1 0 2 = move 0 2
```

2.12.2 Successful Moves

The sequence of moves for solving the Tower of Hanoi with 5 disks from peg 0 to peg 2:

```
0 → 2
0 → 1
2 → 1
0 → 2
1 → 0
1 → 2
0 → 2
0 → 1
2 → 1
2 → 0
1 → 0
2 → 1
0 → 2
0 → 1
2 → 1
0 → 2
1 → 0
1 → 2
0 → 2
1 → 0
2 → 1
2 → 0
1 → 0
1 → 2
0 → 2
0 → 1
2 → 1
0 → 2
1 → 0
1 → 2
0 → 2
```

2.12.3 Verirxefication of Moves

The moves listed above were tested on the Tower of Hanoi simulation website and confirmed to successfully find the solution for moving 5 disks from peg 0 to peg 2.

Question: Would the Tower of Hanoi problem still have a recursive definition if there were infinitely many disks?

2.13 Week 13: HW 13

2.13.1 Exercise 2: Lambda Calculus Interpreter Testing

Lambda expressions from the lectures on Lambda Calculus and Church Encodings were added to `test.lc` and tested with the interpreter.

Test Results:

- `a b c d` → `(((a b) c) d)` (left associativity confirmed)
- `(a)` → `a` (parentheses reduction)
- `(λx.x) a` → `a` (beta-reduction working correctly)
- HW5 example: `(λf.λx.f(f(x))) (λf.λx.(f(f(f x))))` evaluates correctly

All tests confirmed the interpreter conforms to lambda calculus operational semantics.

2.13.2 Exercise 3: Capture-Avoiding Substitution

Investigation of capture-avoiding substitution through test cases and source code analysis.

Implementation:

The `substitute` function (lines 70-87) handles capture-avoidance by generating fresh variable names when substituting into lambda abstractions. When substituting into `λy.e` where the bound variable differs from the substitution target, the implementation:

- Generates a fresh name (Var1, Var2, etc.) via `NameGenerator`
- Alpha-renames the lambda abstraction
- Performs substitution in the renamed body

Test Case:

`(λy.x) [y/x]` → `(λVar1.y)` demonstrates that the free variable `x` is replaced by `y`, while the lambda is renamed to `Var1` to avoid variable capture. If the bound variable matches the substitution target, no substitution occurs (lines 78-79), preserving the binding.

2.13.3 Exercise 4: Normal Form Reduction

Investigation of whether all computations reduce to normal form.

Findings:

Not all computations reduce to normal form. The omega combinator `(λx.x x) (λx.x x)` causes infinite recursion (RecursionError), demonstrating that non-terminating expressions exist.

Evaluation Strategy:

The interpreter uses call-by-name evaluation (lines 37-53): it evaluates the left side of applications first, then substitutes without evaluating arguments.

- Terminating expressions reach normal form
- Non-terminating expressions loop indefinitely
- Expressions already in normal form (e.g., `x y` where variables are free) remain unchanged

2.13.4 Exercise 5: Minimal Non-Terminating Expression

Search for the smallest lambda expression that does not reduce to normal form.

Minimal Working Example:

The omega combinator `(λx.x x) (λx.x x)` is the minimal non-terminating expression.

Analysis:

- Smaller expressions like `x`, `λx.x`, `x x`, or `λx.x x` are all in normal form and terminate
- The omega combinator applies a self-applicative lambda `λx.x x` to itself
- When evaluated: `(λx.x x) (λx.x x)` → substitute `(λx.x x)` for `x` in `x x` → `(λx.x x) (λx.x x)` again, creating infinite recursion

This is minimal because it requires both a self-applicative lambda and applying that lambda to itself, which is exactly what the omega combinator provides.

2.13.5 Exercise 7: Step-by-Step Evaluation Trace

Tracing the evaluation of `((\m.\n. m n) (\f.\x. f (f x))) (\f.\x. f (f (f x)))` following the interpreter's exact steps.

Evaluation Trace:

1. `((\m.\n. m n) (\f.\x. f (f x))) (\f.\x. f (f (f x)))`
2. `((\Var1. (\f.\x. f (f x)) Var1)) (\f.\x. f (f (f x)))`
   - Substituting `(\f.\x. f (f x))` for `m` in `(\n. m n)`
   - The lambda variable `n` is renamed to `Var1` to avoid capture
   - Result: `(\Var1. (\f.\x. f (f x)) Var1)`
3. `((\f.\x. f (f x)) (\f.\x. f (f (f x))))`
   - Substituting `(\f.\x. f (f (f x)))` for `Var1` in `((\f.\x. f (f x)) Var1)`
   - Result: application of Church numeral 2 to Church numeral 3
4. `(\Var5. ((\f.\x. f (f (f x))) ((\f.\x. f (f (f x))) Var5)))`
   - Evaluating the application: `(\f.\x. f (f x)) (\f.\x. f (f (f x)))`
   - Substituting Church numeral 3 for `f` in `(\x. f (f x))`
   - The lambda variable `x` is renamed to `Var5` to avoid capture
   - Final result: partially applied function (not fully reduced to Church numeral 9)

Observations:

The interpreter uses call-by-name evaluation, so it doesn't fully reduce nested applications. The result shows a partially evaluated expression rather than the Church numeral 9, demonstrating that the evaluation strategy affects the final form.

2.13.6 Exercise 8: Recursive Call Trace

Tracing recursive calls to `evaluate()` and `substitute()` for `((\m.\n. m n) (\f.\x. f (f x))) (\f.\x. f x)` with indentation showing the call stack.

Recursive Trace:

```
37: eval (((\m.(\n.(m n))) (\f.(\x.(f (f x))))) (\f.(\x.(f x))))
    39: eval ((\m.(\n.(m n))) (\f.(\x.(f (f x)))))
        39: eval (\m.(\n.(m n)))
    44: subst (\n.(m n)) [(\f.(\x.(f (f x))))/m]
        85: subst (m n) [Var1/n]
            85: subst m [Var1/n]
            85: subst n [Var1/n]
        85: subst (m Var1) [(\f.(\x.(f (f x))))/m]
            85: subst m [(\f.(\x.(f (f x))))/m]
            85: subst Var1 [(\f.(\x.(f (f x))))/m]
        39: eval (\Var1.((\f.(\x.(f (f x)))) Var1))
    44: subst ((\f.(\x.(f (f x)))) Var1) [(\f.(\x.(f x)))/Var1]
        85: subst (\f.(\x.(f (f x)))) [(\f.(\x.(f x)))/Var1]
            85: subst (\x.(f (f x))) [Var2/f]
                85: subst (f (f x)) [Var3/x]
                    85: subst f [Var3/x]
                    85: subst (f x) [Var3/x]
                        85: subst f [Var3/x]
                        85: subst x [Var3/x]
                85: subst (f (f Var3)) [Var2/f]
                    85: subst f [Var2/f]
                    85: subst (f Var3) [Var2/f]
                        85: subst f [Var2/f]
                        85: subst Var3 [Var2/f]
            85: subst (\Var3.(Var2 (Var2 Var3))) [(\f.(\x.(f x)))/Var1]
                85: subst (Var2 (Var2 Var3)) [Var4/Var3]
                    85: subst Var2 [Var4/Var3]
                    85: subst (Var2 Var3) [Var4/Var3]
                        85: subst Var2 [Var4/Var3]
                        85: subst Var3 [Var4/Var3]
                85: subst (Var2 (Var2 Var4)) [(\f.(\x.(f x)))/Var1]
                    85: subst Var2 [(\f.(\x.(f x)))/Var1]
                    85: subst (Var2 Var4) [(\f.(\x.(f x)))/Var1]
                        85: subst Var2 [(\f.(\x.(f x)))/Var1]
                        85: subst Var4 [(\f.(\x.(f x)))/Var1]
        85: subst Var1 [(\f.(\x.(f x)))/Var1]
    39: eval ((\Var2.(\Var4.(Var2 (Var2 Var4)))) (\f.(\x.(f x))))
        39: eval (\Var2.(\Var4.(Var2 (Var2 Var4))))
    44: subst (\Var4.(Var2 (Var2 Var4))) [(\f.(\x.(f x)))/Var2]
        85: subst (Var2 (Var2 Var4)) [Var5/Var4]
            85: subst Var2 [Var5/Var4]
            85: subst (Var2 Var4) [Var5/Var4]
                85: subst Var2 [Var5/Var4]
                85: subst Var4 [Var5/Var4]
        85: subst (Var2 (Var2 Var5)) [(\f.(\x.(f x)))/Var2]
            85: subst Var2 [(\f.(\x.(f x)))/Var2]
            85: subst (Var2 Var5) [(\f.(\x.(f x)))/Var2]
                85: subst Var2 [(\f.(\x.(f x)))/Var2]
                85: subst Var5 [(\f.(\x.(f x)))/Var2]
        39: eval (\Var5.((\f.(\x.(f x))) ((\f.(\x.(f x))) Var5)))
```

Observations:

- Line 37: Entry point to `evaluate()`

- Line 39: Recursive calls to `evaluate()` for left side and result evaluation
- Line 44: Calls to `substitute()` from `evaluate()` for beta-reduction
- Line 85: Recursive calls to `substitute()` for applications and nested substitutions
- Indentation shows the call stack depth, similar to the Tower of Hanoi trace
- The trace demonstrates the nested nature of capture-avoiding substitution, with multiple levels of variable renaming (Var1, Var2, Var3, Var4, Var5)

Question: Are there lambda expressions that have normal forms but no algorithm can find them?

3 Essay

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, Title, Publisher, Year.